



ISLAMIC UNIVERSITY OF TECHNOLOGY

Lab Report 5

Livestock Management System

CSE 4408: System Analysis and Design Lab

Lab 5: Information Gathering: Interactive Methods

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1. Introduction

The purpose of this report is to present the information-gathering activities for the proposed Livestock Management System. The system is designed for livestock farms that currently depend heavily on manual records, disconnected workflows, and delayed reporting. These limitations reduce visibility of farm operations and make decision-making slower for owners, managers, veterinarians, and workers.

The project focuses on identifying user needs through interactive elicitation methods. The selected methods include stakeholder interviews, questionnaires, storytelling, requirement elicitation analysis, and a feasibility review of a Joint Application Development (JAD) workshop. The findings are used to derive, prioritize, and justify high-level system requirements.

1.1 Project Overview

Current Issues	Proposed System Direction
Manual record keeping	Digital and centralized record management
Disconnected subsystems	Integrated livestock, health, finance, and inventory modules
Poor resource tracking	Inventory and feed consumption monitoring
Inefficient reporting	Automated report generation
Delayed decision making	Real-time dashboards and analytics
No centralized monitoring	Role-based access to centralized system data

1.2 Scope of This Report

- Identify the key stakeholder groups and their role descriptions.
- Define interview objectives and prepare interview guides.
- Summarize interview findings and extract requirements.
- Design a questionnaire and present simulated response results.
- Prioritize system requirements based on urgency, value, and feasibility.
- Discuss storytelling and JAD workshop feasibility as additional elicitation methods.

2. Stakeholder Interview Planning and Reports

2.1 Key Stakeholder Groups and Role Descriptions

Stakeholder Group	Role Description	Why This Group Was Selected
Farm Owner	Responsible for business goals, profitability, investment decisions, and expansion planning.	The owner needs financial visibility and reliable reports for long-term planning.
Farm Manager	Coordinates daily operations, resource allocation, inventory, worker schedules, and reporting.	The manager understands operational workflows and the problems caused by manual coordination.
Farm Workers	Record daily activities such as feeding, health observations, and routine tasks.	Workers are the primary data entry users and can reveal usability problems.
Veterinarian	Handles health monitoring, vaccination records, disease treatment, and medical history.	Veterinary input is essential for health tracking and alert requirements.

2.2 Interview Objectives

Stakeholder	Interview Objectives
Farm Owner	Understand business goals, profitability concerns, reporting needs, investment uncertainty, and expansion planning.
Farm Manager	Understand daily workflow, resource allocation, inventory control, scheduling, reporting pain points, and central overview needs.
Farm Workers	Understand daily data entry tasks, record keeping difficulties, mobile usability needs, and clarity of instructions.
Veterinarian	Understand health monitoring process, vaccination records, treatment history, disease management, and late case reporting problems.

2.3 Interview Opening Script

Good morning/afternoon. We are students working on a Livestock Management System for our System Analysis and Design Lab. The purpose of this interview is to understand your current work process, problems, and expectations from a digital livestock management system. Your responses will be used only for academic analysis. There are no right or wrong answers. We may ask follow-up questions to understand your experience better. This interview should take about 15-20 minutes.

2.4 Interview Structure and Questioning Approach

Different interview structures were selected based on the stakeholder type. For farm workers and managers, the pyramid structure is useful because it starts with specific and easier questions before moving to broader open-ended discussion. For farm owners and veterinarians, the funnel structure is useful because they are likely to provide broader priorities first and then move into specific details. The diamond structure may be used when both exploration and confirmation are required.

Structure	Flow	Best Used For	Reason
Pyramid	Specific to general; closed to open-ended	Farm workers and managers	Helps warm up hesitant interviewees and gradually develops deeper discussion.
Funnel	General to specific; open-ended to closed	Farm owners and veterinarians	Allows knowledgeable stakeholders to explain priorities before details are confirmed.
Diamond	Specific to general to specific	Mixed or follow-up sessions	Combines exploration with final confirmation of important facts.

2.5 Closing Script

Thank you for your time and valuable input. Your feedback helps us identify the real requirements of the Livestock Management System. We will summarize the key points and may contact you again if we need clarification. The next step is to compare findings from different stakeholders and prepare a prioritized list of system features.

2.6 Interview Report Summary

Stakeholder	Brief Context	Key Findings	Notable Quote / Insight
Farm Owner	Simulated interview focused on business visibility and decision-making.	Owner needs profit visibility, financial summaries, livestock performance reports, and confidence for investment decisions.	"I cannot plan expansion properly because I do not always know the real cost and performance of each animal group."
Farm Manager	Simulated interview focused on daily operation and coordination.	Manager faces operational overload, inventory uncertainty, scheduling issues, and lack of a central overview.	"I spend too much time checking notebooks and asking workers for updates."
Farm Worker	Simulated interview focused on daily record entry and usability.	Workers experience delayed entries, missing records, unclear instructions, and need mobile-friendly data entry.	"During busy periods, records are delayed or missed."
Veterinarian	Simulated interview focused on animal health and treatment history.	Veterinarian needs complete medical history, vaccination scheduling, treatment records, and disease alerts.	"Late reporting makes treatment harder and increases the chance of disease spreading."

2.7 Common Issues Found Across Stakeholders

- Poor record keeping: data is stored manually and is difficult to retrieve accurately.
- Lack of real-time data: decisions are delayed because live updates are unavailable.
- Communication gaps: farm owner, manager, workers, and veterinarian do not share the same updated information.

- Manual and time-consuming processes: repetitive tasks reduce productivity and increase the chance of error.

3. User Questionnaire Design and Results

3.1 Questionnaire Purpose, Confidentiality, and Instructions

The questionnaire was designed to collect structured feedback from a wider group of livestock system users. The purpose was to validate issues found in interviews, measure user readiness for digital tools, and identify desired features. Responses are treated as confidential and used only for academic requirement analysis. Participants were instructed to answer honestly based on their actual work experience.

3.2 Target Respondent Profile

Respondent Type	Reason for Inclusion	Delivery Method	Reminder Strategy
Farm Owner	Provides business and financial priorities.	In-person or online form.	One polite follow-up after 2 days.
Farm Manager	Provides operational and inventory details.	In-person form or interview-assisted form.	Same-day reminder during work break.
Farm Worker	Provides practical usability and daily data entry feedback.	Assisted paper or mobile form.	Verbal reminder from manager.
Veterinarian	Provides health management and alert requirements.	Online form or short scheduled session.	Follow-up message after 3 days.

3.3 Questionnaire Items

The questionnaire used a mix of closed-ended questions and two open-ended questions. Closed-ended questions used Yes/No, multiple choice, and 5-point rating options. Wording was kept neutral to avoid leading respondents.

No.	Question	Question Type	Response Options
1	What is your role in the farm?	Closed	Owner / Manager / Worker / Veterinarian / Other
2	How are livestock records currently	Closed	Notebook / Spreadsheet / Software /

	maintained?		Mixed / Not sure
3	How often are records updated?	Closed	Daily / Weekly / Monthly / Only when needed
4	How difficult is it to find old records?	Closed	Very easy / Easy / Neutral / Difficult / Very difficult
5	Do you need real-time dashboard information?	Closed	Yes / No / Not sure
6	Which feature is most important to you?	Closed	Inventory / Health tracking / Financial reports / Alerts / Mobile entry
7	How comfortable are you using a mobile app for farm tasks?	Closed	Very comfortable / Comfortable / Neutral / Uncomfortable / Very uncomfortable
8	Do you need reminders for vaccination, feed stock, or follow-up treatment?	Closed	Yes / No / Not sure
9	What is the biggest problem in the current process?	Open-ended	Free text
10	What improvement would help you the most?	Open-ended	Free text

3.4 Survey Participation Summary

Respondent Group	Targeted	Responded	Response Rate
Farm Owner	2	2	100%
Farm Manager	4	3	75%
Farm Worker	10	8	80%
Veterinarian	3	2	67%
Total	19	15	79%

3.5 Closed-Ended Response Summary

Question Area	Main Result	Interpretation
Current record method	Most respondents reported notebook-based or mixed manual records.	Manual record keeping is still the dominant process.
Record update frequency	Most updates are daily, but some are delayed during busy periods.	The system must support quick and simple daily entry.

Difficulty finding records	Majority selected difficult or very difficult.	Searchable digital records are a high-priority need.
Need for real-time dashboard	Most respondents selected yes.	Real-time monitoring should be included in the core system.
Most important feature	Inventory, health tracking, and financial reports received the highest interest.	Priority should be given to operational, health, and reporting modules.
Mobile comfort	Most workers were neutral to comfortable, but some need support.	The interface must be simple, local-language friendly, and mobile accessible.
Reminder need	Most respondents selected yes.	Automated alerts and reminders should be added for vaccination and inventory.

3.6 Thematic Synopsis of Open-Ended Feedback

Theme	Summary of Feedback	Requirement Implication
Manual records	Respondents said notebooks are hard to search and can be incomplete.	Centralized digital records with search and filters.
Late updates	Busy work periods cause missing or delayed entries.	Mobile-friendly quick entry and automated reminders.
Communication gaps	Different people often hold different versions of information.	Shared dashboard and role-based access.
Health risks	Incomplete medical history can lead to wrong treatment or delayed response.	Treatment history, vaccination schedule, and disease alerts.
Financial uncertainty	Owners need clearer profit and cost visibility.	Financial summaries and livestock performance reporting.

4. Requirements Prioritization Planning

Requirements were prioritized by considering stakeholder importance, operational urgency, frequency of use, feasibility, and expected impact on decision-making. Requirements that solve daily operational problems and reduce risk were marked high priority. Requirements that

improve analysis and customization were marked medium priority. Advanced or future-facing features were marked low priority.

4.1 Prioritized Requirement List

Priority	Requirement	Source Objective	Justification
High	Inventory Management	Manager and worker interviews; questionnaire results	Inventory is used daily and directly affects feed availability and farm operations.
High	Automated Report Generation	Owner and manager interviews	Reports are needed for financial summaries, livestock performance, and inventory status.
High	Real-Time Dashboard	Owner, manager, and questionnaire results	Real-time visibility reduces delayed decision-making and improves centralized monitoring.
High	Health and Vaccination Tracking	Veterinarian interview	Health history and vaccination schedules reduce treatment mistakes and late case reporting.
High	Financial Analytics	Owner interview	Profit visibility and cost tracking support business planning and investment decisions.
Medium	Production Performance Analysis	Owner and manager objectives	Useful for comparing productivity but can be added after core records are stable.
Medium	Supplier Management	Manager objective	Improves purchasing control but is less urgent than inventory tracking.
Medium	Data Visualization Dashboard	Owner and manager feedback	Helpful for presentation and analysis after accurate data collection exists.
Medium	Notification and Alert Customization	Veterinarian and manager feedback	Important, but customization can be developed after basic reminders.
Medium	Mobile-Friendly Data Entry	Worker feedback	Important for usability; should be included in design, with advanced features added later.
Low	IoT Sensor Integration	Future automation option	Valuable but requires extra hardware and cost.
Low	Marketplace Integration	Future business expansion option	Not essential for solving current record and monitoring problems.
Low	Predictive Disease Detection	Advanced analytics option	Requires large and reliable historical data.
Low	AI-Based Recommendations	Future intelligence feature	Useful later after sufficient data is collected.

Low	Third-Party System Integration	Future expansion option	Can be considered after the core system is stable.
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4.2 Requirement Categories

Category	Requirements
Operational Requirements	Centralized livestock records, daily feed consumption tracking, inventory monitoring.
Health Management Requirements	Vaccination scheduling, treatment history maintenance, disease outbreak alerts.
Reporting Requirements	Financial summaries, livestock performance reports, inventory status reports.
User Requirements	Role-based access control, mobile accessibility, easy-to-use interface, local language interface.

5. Other Interactive Elicitation Methods

5.1 Storytelling Analysis

Storytelling is useful because users often explain hidden requirements through real work situations rather than direct feature requests. A story can reveal exceptions, delays, communication problems, and workarounds that may not appear in a normal questionnaire.

Example worker story: “I record feed usage and animal health information in a notebook every day. During busy periods, some records are delayed or missed. When the manager requests information, finding the correct records takes time and sometimes important details are overlooked.”

Story Element	Requirement Extracted
Records are written in notebooks.	Digital record management and centralized database.
Entries are delayed or missed during busy periods.	Mobile-friendly data entry and automated reminders.
Finding records takes time.	Searchable records and filters.

Important details are overlooked.	Real-time reporting and validation checks.
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5.2 JAD Workshop Feasibility Assessment

A Joint Application Development workshop can help stakeholders discuss requirements together, resolve misunderstandings, and agree on priorities. However, full-scale JAD may be difficult because livestock stakeholders may have busy schedules, multiple farm locations, and different availability.

Factor	Assessment
Scheduling constraints	High. Owners, managers, workers, and veterinarians may not be available at the same time.
Cultural readiness	Moderate. Some users may be unfamiliar with structured workshops but can participate in guided discussion.
Complexity	Moderate. The system includes operational, health, inventory, finance, and reporting modules.
Expected benefit	High. A workshop can align owner, manager, and veterinarian expectations.
Recommended approach	A small-scale JAD session involving owner, manager, and veterinarian during detailed analysis.

5.3 JAD Conclusion

A small-scale JAD session is advisable. It should include the farm owner, farm manager, and veterinarian first because these participants can define core business, operational, and health management requirements. Worker feedback should be collected separately through assisted interviews or mobile usability testing, because workers may be more comfortable giving practical feedback in a smaller setting.

6. Conclusion

The information-gathering activities show that a Livestock Management System is needed to reduce manual work, improve record accuracy, support health monitoring, and provide real-time operational visibility. Interviews revealed that different stakeholders face different problems, but

several issues are common: poor record keeping, lack of real-time data, communication gaps, and manual repetitive processes.

The questionnaire supported the interview findings and showed strong interest in inventory management, health tracking, automated reporting, mobile data entry, and reminders. The highest priority requirements are inventory management, automated report generation, real-time dashboard, health and vaccination tracking, and financial analytics. Advanced features such as IoT integration, AI recommendations, and predictive disease detection should be considered only after the core system is stable and reliable data has been collected.

Appendix A: Interview Guide

A.1 Farm Manager Interview Guide

- Describe a typical workday on the farm.
- What tasks consume most of your time?
- What problems occur most frequently?
- Do you currently maintain digital records?
- Are inventory records updated daily?
- How do you maintain worker shift scheduling?
- Why does that issue occur?
- How often does it happen?
- What would improve this process?

A.2 Farm Owner Interview Guide

- What business goals should the system support?
- What reports do you need most frequently?
- How do you currently evaluate profitability?
- What information is missing when making investment decisions?
- What kind of dashboard would help you plan expansion?

A.3 Veterinarian Interview Guide

- How do you currently review animal medical history?

- What vaccination information must be recorded?
- What problems occur because of late case reporting?
- What alerts would help you respond faster?
- How should treatment history be maintained?

Appendix B: Questionnaire

Instruction: Please answer the following questions based on your current farm management experience. Your responses will remain confidential and will be used only for academic requirement analysis.

1. What is your role in the farm? Owner / Manager / Worker / Veterinarian / Other
2. How are livestock records currently maintained? Notebook / Spreadsheet / Software / Mixed / Not sure
3. How often are records updated? Daily / Weekly / Monthly / Only when needed
4. How difficult is it to find old records? Very easy / Easy / Neutral / Difficult / Very difficult
5. Do you need real-time dashboard information? Yes / No / Not sure
6. Which feature is most important to you? Inventory / Health tracking / Financial reports / Alerts / Mobile entry
7. How comfortable are you using a mobile app for farm tasks? Very comfortable / Comfortable / Neutral / Uncomfortable / Very uncomfortable
8. Do you need reminders for vaccination, feed stock, or follow-up treatment? Yes / No / Not sure
9. What is the biggest problem in the current process?
10. What improvement would help you the most?

References

Team Go Paloks. Livestock Management System, CSE 4408: System Analysis and Design Lab, Presentation 5, June 03, 2026.